
SUMMARY

Machine Learning Engineer and AI researcher with 6+ years of experience developing scalable, decision-support systems that integrate LLM agents, computer vision, Bayesian modeling, and uncertainty-aware scientific machine learning. Experienced in building production-oriented ML workflows using Python, PyTorch, FastAPI, Docker, and AWS, with applications across architectural drawing intelligence, building energy analytics, retrofit decision support, and complex engineering systems.

TECHNICAL SKILLS

ML Frameworks: PyTorch, TensorFlow, Hugging Face, LangChain, scikit-learn, JAX, PyMC, Pyro
Languages: Python, C/C++, MATLAB, SQL, Java, R
Infrastructure: Git, Linux, Azure, Docker, AWS, PostgreSQL, REST APIs
Methods: LLM, Agentic AI, Physics-Informed Machine Learning, Scientific ML, Bayesian Inference, Uncertainty Quantification, Causal Discovery and Inference, Model Selection, Digital Twins, Equitable Decision Making

EXPERIENCE

AI Software Engineering Intern May 2026 – Aug. 2026
Builder Supply Link NJ, USA

- **Computer Vision:** Developed a YOLO-, LayoutLM-, and CubiCasa5K-assisted pipeline for architectural drawing analysis, extracting symbols, dimensions, metadata, room geometry, and product-relevant information.
- **Full-Stack Development:** Developed asynchronous Python/FastAPI backend services with React/Next.js front-end interfaces for architectural document understanding and cost estimations.
- **Product Recommendation:** Integrated RAG workflows with BigCommerce Catalyst and AWS deployment to connect extracted drawing information with actionable product recommendations.

Graduate Research Assistant Aug. 2019 – Present
The Pennsylvania State University University Park, PA, USA

- **Agentic AI Frameworks:** Architected and deployed an LLM multi-agent system for retrofit decision support, orchestrating specialized agents over 140K+ housing records to chain energy diagnostics, building-code retrieval, and automated report generation into a single explainable pipeline.
- **ML-based Diagnostics:** Built large-scale ML pipelines over national energy data, combining ML models, distributional analysis, and an explainability layer, to identify the top drivers of energy inefficiency across U.S. climates.
- **Uncertainty Quantification:** Developed uncertainty-aware workflows using deterministic and stochastic methods to improve forecast reliability and decision confidence.
- **Bayesian Modeling & Inference:** Built Bayesian predictive models with posterior estimation and uncertainty propagation to deliver calibrated, decision-ready predictions for engineering systems.
- **Physics-Informed ML:** Developed physics-guided Bayesian neural networks that embed governing equations into machine learning models to improve prediction reliability, generalization, and uncertainty estimation.
- **Data Engineering:** Created scalable data pipelines that integrate climate, weather-zone, socioeconomic, building-physics, and energy-consumption features into model-ready datasets for retrofit targeting.
- **Bayesian Model Calibration & Selection:** Evaluated Bayesian and information-theoretic model selection methods to balance model fit, complexity, and generalization in calibrated predictive models.
- **Causal Inference:** Applied causal discovery, causal inference, knowledge-guided graphs, and LLM-assisted reasoning to support more actionable energy-system decision-making.

EDUCATION

Pennsylvania State University, State College, PA, USA Aug. 2019 – Sept. 2026
Ph.D., Architectural Engineering; Ph.D. Minor in Computational Science

Arizona State University, Tempe, AZ, USA Aug. 2022 – Dec. 2024
M.S., Computer Science, concentration in Machine Learning

Tongji University, Shanghai, China Jul. 2015 – May 2019
B.S., Mechanical Engineering; Outstanding Graduate

HONORS & AWARDS

Outstanding Dissertation Award (\$2500) Jun 2026
The Wormley Family Graduate Fellowship (\$3,000) Jun. 2023
Louis S. and Sara S. Michael Endowed Graduate Fellowship in Engineering (\$3,000) Jun. 2022
University Graduate Fellowship (\$30,000 per year) Aug. 2019 – May 2024
Glenn E. Singley Memorial Graduate Fellowship (\$8,000) Aug. 2019
Outstanding Graduate of Tongji University May 2019